

Algebra 1
Unit 5
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

Marquise wants to save up for a pair of new sneakers that cost \$120. He has already saved \$37.50. He downloaded an app that will allow him to add ten cents to his savings account whenever he pays with his debit card. He uses his debit card to make an average of three purchases per day.

1. Define the variables in this context.

x = number of days

y = total saved

2. What is the parameter?

He earns an average of \$0.30 per day.

3. What is the constant?

37.50

4. Which variable represents the output in this situation?

y

5. Complete the table to write an equation that represents the context.

Total saved	=	# purchases	·	# days	+	Starting balance
y	=	0.30	·	x	+	37.50

6. Solve the equation for the unknown

$$120 = 0.3x + 37.50$$

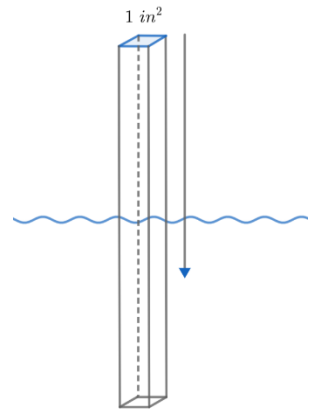
$$82.50 = 0.3x$$

$$x = 275$$

7. Interpret the solution.

It will take 275 days to save the money.

To measure air pressure, scientists consider a 1 in^2 column of air that reaches from sea level all the way up to the atmosphere. Even though air seems “empty”, it actually has weight, which exerts force on objects on the ground! The air applies 14.7 pounds of pressure per square inch (14.7 psi) at sea level.



Water weighs much more than air. The deeper you dive below the surface, the pressure the water exerts. For every foot of depth, the water weighs an additional 0.445 lbs beyond the weight of the 1 in^2 column of air.

8. Define the variables in this context.

$x = \text{depth}$

$y = \text{weight}$

9. Write an equation that can be used to find the pressure exerted on an object in terms of the depth below sea level.

$$y = 0.445x + 14.7$$

10. The human body can take 58 psi of pressure. Use your equation to find the maximum depth under water a human could survive without being crushed by the pressure of the water.

$$58 = 0.445x + 14.7$$

$$43.3 = 0.445x$$

$$x \approx 97.3 \text{ feet}$$